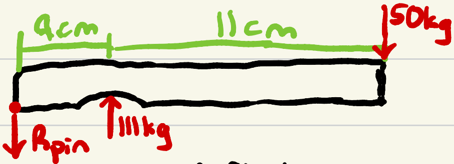
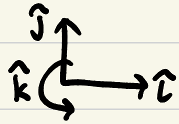
Problem: Build off of the previous macademia nut nutcracker linear actuator design taking into account beam deflection.

a) Find the location of maximum elastic deflection in handle.

Assume: small cut-out for macademia nut doesn't affect maximum elastic deflection.





Top handle deflects most:

$\Sigma F_y = 0 = 111kg - 50kg - R_{pin}$

$R_{pin} = 61kg$

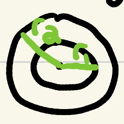
Assume max deflection occurs in the long part of the handle, and model as fixed-free:

Beam Deflection equation for max deflection if fixed-free beams (Appendix E):

$$\boxed{y_{max} = \frac{-PL^3}{3EI} = \frac{-500N(.11m)^3}{3EI} = \frac{-.2218N\cdot m^3}{EI}}$$

b) Choose a beam design so $y_{max} < 2\% L$ and the handle is as mass efficient as possible.

Use hollow tube for ergonomical grip that is mass efficient and strong:



$I = \frac{\pi}{4}(r_2^4 - r_1^4)$ $\boxed{r_2 = 1.25cm}$ based on hand comfort and macademia nut size

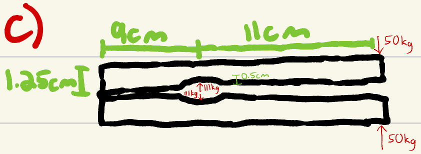
Use Stainless Steel AISI 302 Cold-rolled for high strength, low weight and relative cost, corrosion resistance, and food safe properties: $E = 190\,GPa$

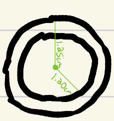
$.02(.11m) = .0022m = \frac{.2218N\cdot m^3}{(190GPa)I}$ $I = 5.306\times10^{-10} = \frac{\pi}{4}(.0125_m^4 - r_1^4)$

$r_1 = 0.0124m = 1.24cm$

Use $\boxed{r_1 = 1.20cm}$ due to assumption oversights.

c)





Stainless steel AISI 302 Cold-Rolled: $E = 190\,GPa$